THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which can conduct electricity? 1

(a) dry paper **(b)** dry wood **(c)** plastic **(d)** salt solution

2. A fuse wire: 1

(a) is a switch **(b)** never melts

(c) melts when the current is too high **(d)** is a magnet

3. A wire is covered with plastic to: 1

(a) make it a magnet **(b)** avoid shocks **(c)** make it heavier **(d)** reduce light

4. Overloading is caused by: 1

(a) a switch **(b)** a thick wire

(c) a fuse **(d)** too many appliances on one socket

5. Heating effect of current is used in: 1

(a) electric iron **(b)** magnet **(c)** compass **(d)** cell

6. A bulb glows due to the: 1

(a) heating of its filament **(b)** sound

(c) magnet **(d)** water

7. A short circuit can cause: 1

(a) fire **(b)** a brighter glow in all bulbs

(c) a stronger magnet **(d)** a fall in current

8. Filament of a bulb is made of: 1

(a) rubber **(b)** tungsten **(c)** iron only **(d)** wood

9. MCB means: 1

(a) miniature circuit breaker **(b)** magnetic coil bulb

(c) main current bulb **(d)** motor circuit battery

10. Electromagnet loses magnetism when current is: 1

(a) doubled **(b)** never **(c)** switched off **(d)** switched on

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Which effect of current is used in a geyser? 2

12. What is an electromagnet? 2

13. Name the metal used in the bulb filament and the heating element. 2

14. Why are wires coated with plastic? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State three uses of heating effect of current. 3

16. Why is a fuse used in series? 3

17. What happens to an electromagnet when the current is switched off? 3

OR

Why does a bulb glow? 3

18. Explain the open and closed circuit. 3

OR

Why should we not touch a switch with wet hands? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the heating effect of current and its use. 5

OR

Write about short circuit and overloading. 5

20. Write about a fuse and the reasons it is used. 5

OR

Differentiate between a bulb filament and a heater coil. 5

END OF QUESTION PAPER — Total: 40 Marks

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